

Stochastic Wage Suppression on Gig Platforms and How to Organize Against It

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Abstract

Digital labor markets are increasingly used to procure diverse forms of human input, from data annotation to food delivery. A central concern in such markets is the ability of platforms to suppress wages by exploiting the abundance of low-cost labor. To better understand labor outcomes, we introduce a novel posted-price procurement model with coverage objectives. In our model a platform seeks to acquire a set of M tasks while minimizing wait time, as well as total spending. Workers are sampled from a population and complete a task if the posted price is higher than their estimated cost of labor. First, we show that with a simple pricing strategy the platform can cover all categories in time $O(M)$, while paying only a $O(\log(M)/M)$ fraction of the total cost of labor by exploiting high worker uncertainty about their costs. Then, we study the impact of collective action to prevent the exploitation of workers. We show how a small, *strategically chosen* coalition of workers that commits to a price floor forces the platform's total spending from logarithmic to linear in M , substantially reducing the power of the platform to suppress wages. In contrast, a randomly sampled coalition of equal size remains largely ineffective. We complement our theory with synthetic experiments that demonstrate the benefit of targeted recruitment for collective action across different market regimes. More broadly, our results provide a theoretical foundation for understanding how to organize collective bargaining for promoting welfare in digital labor markets.

CCS Concepts

• **Applied computing** → **Online auctions**; • **Mathematics of computing** → *Stochastic processes*; *Distribution functions*; • **Theory of computation** → *Probabilistic computation*.

Keywords

online gig economy, collective action, posted-price mechanism

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1 Introduction

The rise of the gig economy has changed the way people get paid for work [52]. On the one hand, gig platforms increasingly fragment work into hundreds of fine-grained microtasks—from labeling data and verifying outputs of AI models to moderating online content (e.g., on Amazon Mechanical Turk or Upwork). On the other hand, employers can post wages and dynamically adapt their pricing to worker supply: for example, ride-sharing platforms such as Uber or Lyft employ algorithmically chosen prices to facilitate rides between drivers and customers. Workers accept tasks when the posted wage exceeds their privately estimated cost of completion. Empirical research on gig platforms shows that workers exhibit high uncertainty around their own costs [24] and often overlook the hidden costs of searching, monitoring, and payment management [15, 53].

A growing concern is that platforms can exploit this uncertainty, as well as the diversity of tasks, to design pricing schemes that result in under-market wages [24] and may even cause workers to operate at a loss [42]. Dubal [18] argues that employers on gig platforms are able to “use granular data to produce unpredictable, variable, and personalized hourly pay.” While empirical research documents wage variability and underpayment, the theoretical foundations linking pricing strategies to wage suppression remain underexplored.

Collective action—through coordinated price floors, information campaigns, or shared organizing tools—has therefore emerged as one of the few levers available to workers [28, 48, 49, 60]. Our work provides a formal model that explains when and how such coordination can counteract algorithmic wage suppression. We provide a theoretical framework studying wage suppression on gig platforms and design interventions that mitigate it through targeted recruitment of low-cost workers.

1.1 Our contributions

We propose a novel procurement model in which a principal—e.g., a firm or a platform—wishes to obtain the completion of tasks across M distinct categories. These categories might represent different types of data annotation (e.g., labeling images, annotating medical data) or tasks such as red-teaming exercises for machine-learning models.

The process unfolds in iterations. At the start, all M categories are active. At iteration t , the principal posts a price p_t that applies to any task from the currently active categories. Workers have a true cost associated with each task category, for which they have a noisy estimate, denoted as their *valuation*. Workers arrive sequentially and draw their valuations from a set of distributions, one for each task category. These distributions model the workers' uncertainty in

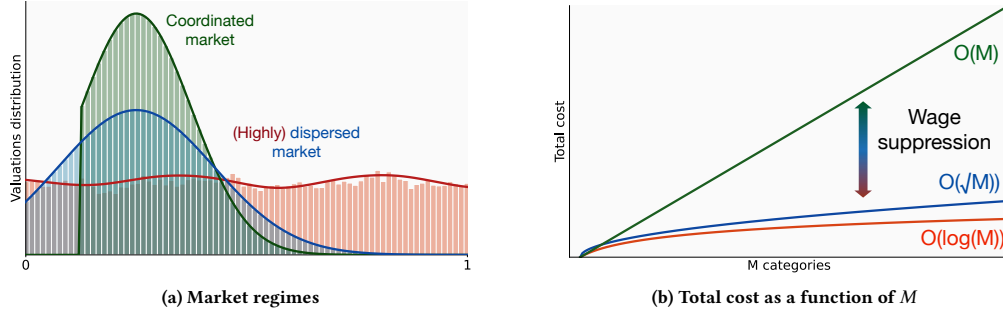


Figure 1: (a) Dispersed markets (blue and red) are emblematic of workers’ high uncertainty over their own costs and presence of workers willing to take arbitrarily low prices for tasks, while coordinated markets (green) represent cases where workers have coordinated on a price floor per category. (b) A principal can exploit dispersed markets, reducing his total cost from $O(M)$ to sublinear, even logarithmic or constant with respect to M , while preserving a linear wait time guarantee.

their cost estimate. A worker accepts a task in some category if their valuation for that category is at most the posted price. The workers behave rationally by only accepting tasks that have a positive profit.

A **success event** occurs when a worker accepts a task. In this case, the principal pays the posted price p_t , and the corresponding category is removed from the active set. The process continues with the reduced set of categories and a (possibly updated) price p_{t+1} . The **wait time** for iteration t is the number of workers that had to be drawn until the first success.

The principal’s objective is to cover all M categories—i.e., to elicit at least one successful completion in each category—while minimizing both the total wait time across iterations and total payments. If there is a non-negligible price floor p for each category (e.g., the market rate), the principal must pay a total cost of pM to ensure coverage of all categories in finite time, achieving an optimal wait time of M . If such a floor does not necessarily exist (i.e., workers accept a task at arbitrarily low wages), the principal can exploit the *tails* of the valuations distributions and suppress wages while still preserving a wait time of $O(M)$ time with high probability. We denote this as a *stochastic wage suppression* pricing strategy and analyze its implications for workers’ total welfare:

- We characterize necessary properties of the workers’ valuations distributions that lead to different wage suppression regimes. We connect these properties with empirical observations in the gig economy. Figure 1 showcases our main results: workers who have a price floor obtain a total wage of $\Theta(M)$; as their uncertainty increases and low-wage workers appear, the principal can suppress wages down a sublinear regime (even logarithmic or constant as a function of M) with a wait time $O(M)$ with high probability (Section 3);
- We show that untargeted collective action strategies have limited impact: if a random α -fraction of workers coordinate on a price floor in each category, the wage suppression bounds remain unchanged for any fixed collective fraction $\alpha < 1$ (Section 4);
- We show that targeted recruiting restores total wages to a bound of $\Theta(M)$: for any α -fraction of categories, we find a closed-form condition for the minimum budget required to

design a targeted intervention for the workers at the low-end of the valuations distributions. This intervention restores wages from any sublinear regime while maintaining wait time of $O(M)$ with high probability (Section 5).

We show that our theory holds in practical synthetic experiments.¹ Our model is versatile and has minimal assumptions, illustrating how the power of collective bargaining is dependent on the ability of a firm to wait out high-priced workers as well as the workers’ ability to strategically coordinate.

1.2 Related work

Our model is a dual to classical posted-price mechanisms [5, 12, 13, 19, 26, 32], where some works assess monopolistic tendencies of platforms [20, 56]. In our model, instead of a seller maximizing revenue, a buyer (platform) minimizes procurement cost under coverage constraints. In these models, workers exhibit a take-it-or-leave-it behavior based on their valuations and firms optimize to maximize average revenue under full knowledge of valuations distributions.

Our posted-wage setting models truthfully gig economy platforms like Mechanical Turk and Upwork [4, 55] and is closest in spirit to posted-price procurement models [6, 11, 50], but differs in objective and feedback: rather than maximizing the submodular value subject to a budget while learning an estimate of the optimal revenue in hindsight, we optimize the principal’s cost to ensure coverage of all categories. An alternative framework models workers and tasks as a two-sided market with bilateral arrival uncertainty and creates optimal mechanisms for task assignment [17]. We showcase that a simple pricing strategy based on distribution quantiles knowledge allows a principal to suppress prices with bounded guarantees. Quantile information has been recently used in pricing to show that it can lead to robust pricing menus [58]. Moreover, we analyze strategies for collective action that prevent a principal from exploiting low-wage workers, which differs from earlier works.

¹All code used for the experiments and figures is made available online at https://github.com/astoica/pricing_strategies_gig_economy. All experiments use Monte Carlo simulations of various distributions averaged over 60 runs and were run in Python 3.9 on local computers.

Our work is inspired by recent initiatives to organize workers on gig platforms, through information campaigns such as Turkopticon [28], FairWork [23], policy and platform interventions to set minimum wages on driving gig economy [10, 45], and self-organizing campaigns [44, 47, 48, 51, 54] (for a survey see Sigg et al. [49]). Recently, collective action has gained attention in the machine learning literature [c.f., 7, 22, 25, 57, 61]. Most works assume randomly sampled collectives, with recent ones focused on minority collectives [8]. In this context, our work is the first to demonstrate the benefit of targeted recruiting.

2 Market model

We model a simple market in which there is a set of workers $\mathcal{W}$. Each worker is associated with valuations over M categories of data. We model these valuations as the workers' perceived cost for completing tasks in each category. We use f_i to denote the pdf of workers' valuations distribution for category $i \in [M]$, with associated CDF D_i and (left) quantile $Q_i(u) := D_i^{-1}(u) := \inf\{x : D_i(x) \geq u\}$ for $u \in [0, 1]$. We denote $[M] = \{1, 2, \dots, M\}$. Hence, the workers' valuations are i.i.d. draws from a product distribution $\mathcal{D} = D_1 \times \dots \times D_M$ and are assumed to be bounded in $[0, 1]$. We assume the workers behave rationally and accept a task that pays higher than their valuation (they have a positive profit), keeping the option to defer at zero profit.² As such, the valuations are also equivalent to a *reservation price*, i.e., the minimum price for which a worker would accept a task, canonical in models of labor markets [9, 34, 37].

The principal wishes to complete a set A of M tasks. To this end, they post a sequence of prices $\mathbf{p} = (p_1, \dots, p_M)$. For each price $p_t \in \mathbf{p}$, workers arrive sequentially and decide whether to complete one of the remaining tasks $a \in A$ or to defer. The price p_t remains posted until the task is completed. Once a task is completed it is removed from the active set A . The number of deferred events at price p_t is denoted as N_t . This leads to the dynamic outlined in Algorithm 1. We denote by q_t the probability of obtaining a completed task in iteration t , which depends on the pricing strategy and the valuation distributions.

Wait time constraint. A central quantity in our model is the wait time. The *wait time* for the principal, for a sequence of prices $\mathbf{p}$ is

$$\text{WAIT}(\mathbf{p}) = \sum_t N_t. \quad (1)$$

Canonical works in matching markets and labor economics show that price and wait time for task uptake are often at a trade-off [39, 41] and wait time varies as a function of market size [3]. To navigate this, empirical studies of the gig economy also show that employers often impose a wait time constraint through algorithmic means or penalties [31, 36].

In the following we are interested in pricing strategies that guarantee a linear expected wait time in M ,

$$\exists C > 0 : \mathbb{E}[\text{WAIT}(\mathbf{p})] < CM \quad \forall M, \quad (2)$$

where randomness comes from the stochastic arrival process. Under this constraint a principal may care about a cost-time trade-off since it can always achieve the $O(M)$ with a naive pricing strategy. As such, we study the case of large M , motivated by the proliferation

²Alternatively, a worker can choose the category with the largest margin $i^* = \arg\min_{j \in A_t} v_j - p_t$, noting that our results hold since prices are posted *ex-ante*.

Algorithm 1: Platform-worker interaction

Input: Start with $A_1 = [M]$. Define a sequence of prices $\mathbf{p}$:

1 For $t \geq 1$ proceed as follows:

- (1) The principal posts a price p_t for the completion of a task in the set of active categories A_t .
- (2) Repeat until success:
 - (a) A worker is drawn uniformly at random from a distribution of workers. The worker is characterized by a valuation profile v with $v_j \sim D_j, \forall j$ is drawn uniformly at random.
 - (b) A “success” event is defined as

$$S = 1 \left[\min_{j \in A_t} v_j \leq p_t \right].$$

Namely, there is a category j for which the worker's valuation v_j —a proxy for his cost of the task—is lower than the principal's posted price p_t .

- If $S = 1$, the worker completes a task for one of the available categories

$$i^* \in \{j \in A_t : v_j \leq p_t\}$$

- If $S = 0$ a new worker is sampled, independently, and the process repeats from (a).
- (3) The principal pays p_t and the category i^* gets removed from the set of categories that the principal is willing to pay for in the future:

$$A_{t+1} = A_t \setminus \{i^*\}.$$

We advance to iteration $t + 1$ as long as $A_{t+1} \neq \emptyset$. Let N_t count the number of unsuccessful events occurred for step t .

of microtasks in digital crowdwork platforms, whose number and diversity in categories continues to expand [1, 59] as well as the rapid expansion of the gig economy into new market segments [21].

Payment and wage suppression. The principal's *total cost* for a pricing strategy $\mathbf{p}$ is the sum of the prices for all the success events (note that we are guaranteed a success event in each price stage by definition of the model, what varies is the wait time):

$$\text{COST}(\mathbf{p}) = \sum_{t=1}^M p_t \quad (3)$$

We assume that the platform does not take a cut, so the principal's total cost equals the total wages earned by workers. Without much information about the workers' valuations, a principal is limited in their ability to minimize cost and therefore may use a noisy statistic such as pricing at the mean, median, or simply pay the maximum allowed price. Since valuations lie in $[0, 1]$, paying a total cost of M guarantees a wait time of exactly M . By contrast, access to information about valuation distributions allows a price-conscious principal to strategically “wait out” higher-priced workers and therefore pay below-market wages (a strategy observed in markets where employers have significant information about workers' reservation prices or labor supply [2, 10]).

We refer to regimes in which total payments scale sublinearly in M as regimes of *wage suppression*. We show that depending on the shape of the valuation distributions, even partial information—such

as oracle access to quantile values—allows a principal to achieve sublinear total payments, including logarithmic or constant regimes in certain cases.

2.1 Workers' task valuations

In our setup, the workers have valuations that correspond to their perceived cost for completing a task from a certain category. The distribution set $\mathcal{D}$ models the workers' uncertainty over their own costs: for example, if a category i corresponds to annotating images on Mechanical Turk, the workers will estimate their market rate of p dollars per hour for this task category. In our model, the properties of these distributions characterize different market regimes and determine the principal's ability to suppress wages. We discuss some key characteristics of these distributions.

Dispersion. In reality, in many market segments workers have uncertainty over their costs and thus exhibit high dispersion of perceived costs, a phenomenon observed by measuring workers' heterogeneity in reported reservation prices [14]. In online markets, unpaid work is shown to increase workers' uncertainty over their costs [46, 53] and lead to highly dispersed reported bids or accepted wages. We exemplify this case through the uniform distribution $\text{Unif}[0, 1]$. In contrast, a market with no uncertainty over p would be equivalent to setting D_i as the Dirac distribution at p ($v \sim \delta_p$). Highly dispersed markets are vulnerable to wage suppression. A principal is able to exploit the positive mass of low-priced workers (who accept a task at arbitrarily low wages) and benefits from a pricing strategy that pays significantly under the mean or median. We provide conditions for distributions that exhibit dispersion to be exploited. In contrast, more concentrated distributions are more resilient to wage suppression. We exemplify dispersed markets in blue and red in Figure 1.

Right-skew. A prevalent case in online markets features valuations heavily skewed toward zero, i.e., a significant mass of workers accept tasks at arbitrarily low wages. Empirically, reservations prices often exhibit high skewness [34] (often exogenous to the market and influenced by previous wages and time in unemployment). In online markets specifically, wages are often skewed towards the low-earners [24, 27] with reported wage depreciation due to unpaid work. We note that such markets are particularly vulnerable to wage suppression by a principal who strategically sets low prices. We exemplify this case with the Beta($\alpha < 1, 1$) distribution and give a condition based on left-tail properties to show that a principal can achieve a constant total cost as a function of M in such vulnerable market regimes.

Tail properties. In our setting, the relevant behavior for characterizing wage suppression is the distributions' behavior near zero: we characterize the rate of growth of the total cost under strategic pricing as a function of tail growth rate near 0 (i.e., a function characterizing the spread of workers who take arbitrarily low prices). An important special case is when there is a price floor p^* , i.e. a price below which no single worker will perform the task, and thus there is no left tail near 0. When the market value is positive and accurately reflects workers' cost, this essentially imposes a *floor reservation price*, seen in empirical studies of labor markets [34] and collective action [29]. We call these markets *coordinated markets*

and exemplify this case in green in Figure 1. As we will see the existence of a price floor makes markets resilient to wage suppression, limiting a principal's ability to strategically suppress prices while preserving his wait time guarantee.

Next, we analyze a natural and versatile pricing strategy that guarantees linear expected wait time for any valuation distribution. Accordingly, we are interested in characterizing payments in different market regimes by analyzing the limit dependence of the principal's cost on M .

3 Stochastic wage suppression strategy

We propose a general pricing rule that ensure that the expected wait time is $O(M)$ with high probability for any valuation distribution. It is based on evaluating the left quantiles of the distribution, assuming that partial knowledge through oracle access is available to the principal:

Definition 3.1 (Stochastic wage suppression). We define the stochastic wage suppression pricing strategy $\mathbf{p}^{\text{SWS}}$ by setting the posted price at iteration t to the $\theta/|A_t|$ quantile of the workers' valuations distributions:

$$p_t = \max_{i \in A_t} Q_i\left(\frac{\theta}{|A_t|}\right) \quad t = 1, 2, \dots, M, \quad (4)$$

for some $\theta > 0$. In our model, exactly one category is removed at each iteration, so $|A_t| = M - t + 1$ by construction.

We denote this strategy as the *stochastic wage suppression pricing strategy* because it exploits the tails of the valuation distribution to reduce payments for the principal.

3.1 Linear wait time guarantee

Our first main result is that the $\mathbf{p}^{\text{SWS}}$ satisfies the linear expected wait time constraint (2) for any valuation distribution. To show that, we note a sufficient condition that lower-bounds the probability of success for each iteration:

Lemma 3.2 (Lower bound on success probability). *For the stochastic wage suppression pricing strategy $\mathbf{p}^{\text{SWS}}$, there exists a constant $q^* > 0$ independent of M for which the success probabilities satisfy $q_t \geq q^*, \forall t \in [M]$.*

PROOF. We can compute the success probabilities as

$$\begin{aligned} q_t &= 1 - \prod_{i \in A_t} (1 - D_i(p_t)) \geq 1 - \left(1 - \frac{\theta}{M-t+1}\right)^{M-t+1} \\ &\geq 1 - e^{-\theta} =: q^*, \forall t \in [M], \end{aligned} \quad (5)$$

since for each active i , $D_i(p_t) \geq D_i(Q_i(\theta/|A_t|)) = \theta/|A_t|$. By construction, $|A_t| = M - t + 1, \forall t \in [M]$. Independence across categories gives $q_t = 1 - \prod_{i \in A_t} (1 - D_i(p_t)) \geq 1 - (1 - \theta/|A_t|)^{|A_t|} \geq 1 - e^{-\theta}$. Here we used that $(1 - \theta/x)^x \leq e^{-\theta}, \forall x, \theta > 0$. The N_t are Geometric random variables since the process is sequential and each worker is drawn independently, $N_t \sim \text{Geom}(q_t)$ on $\{1, 2, \dots\}$. $\square$

Theorem 3.3 (Wait time). *Consider the stochastic wage suppression strategy $\mathbf{p}^{\text{SWS}}$ defined in Definition 3.1, and let $q_* = 1 - e^{-\theta}$. Then*

$$\mathbb{E}[\text{WAIT}(\mathbf{p}^{\text{SWS}})] \leq \frac{M}{q_*} = O(M). \quad (6)$$

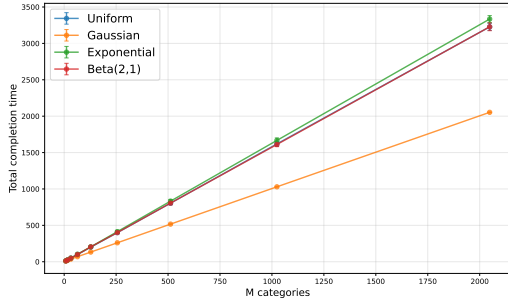


Figure 2: Total wait time for the $\mathbf{p}^{\text{SWS}}$ pricing strategy for different distributions, as M varies.

Moreover, the iteration-wise wait times $\{N_t\}_{t=1}^M$ are independent sub-exponential random variables with parameters depending only on q_* . Hence, for any $\delta \in (0, 1)$,

$$\text{WAIT}(\mathbf{p}^{\text{SWS}}) = O\left(M + \sqrt{M \log(1/\delta)} + \log(1/\delta)\right) \quad (7)$$

with probability at least $1 - \delta$.

PROOF SKETCH. We use a standard result on Geometric random variables that are sub-exponential with ψ_1 -norm bounded by a constant depending only on q_* (shown in Lemma A.1 in the Appendix). We then use the independence of the wait times at every iteration to compute the total wait time in expectation as $\mathbb{E}[\text{WAIT}(\mathbf{p}^{\text{SWS}})] \leq M/q_*$. Finally, we show that the total wait time $\text{WAIT}(\mathbf{p}^{\text{SWS}}) = \sum_t N_t$ satisfies a Bernstein (sub-exponential) tail and we use Bernstein's inequality to prove the wait time concentration bound. $\square$

The $\mathbf{p}^{\text{SWS}}$ strategy has a natural interpretation through quantile access at the inverse number of active categories, which gives the constant lower bound of success in each iteration. When $M = 2$, $\mathbf{p}^{\text{SWS}}$ is pricing at the median for $\theta = 1$; as M increases, access to lower quantile values allows the principal to price under the median while maintaining the wait time. Figure 4 shows the total wait time under the $\mathbf{p}^{\text{SWS}}$ strategy, noting that the $O(M)$ time guarantee holds, with different constants depending on the exact valuations distributions.

3.2 Total payments

While holding the expected wait time fixed, the stochastic wage suppression pricing strategy achieves a different total cost depending on the distribution of valuations among workers. In the following we use the characteristics outlined in Section 2.1 to describe different cost regimes.

Price floors do not induce wage suppression. In coordinated markets where all workers have a minimum positive market price for each category $i \in [M]$ (e.g., there is a minimum wage requirement or there is no uncertainty or dispersion in the workers' costs), the principal pays a total price of $\Theta(M)$. We characterize this property as a sufficient condition on the total cost bound below:

Lemma 3.4 (Linear total cost). *For each category i , let $g_i := \inf\{x \in [0, 1] : D_i(x) > 0\}$ denote the left endpoint of the support (informally,*

the gap at zero price). There is a positive left gap if

$$g_* := \min_{i \in [M]} g_i > 0. \quad (8)$$

For any distributions $\{D_i\}_{i \in [M]}$ with property (8), the $\mathbf{p}^{\text{SWS}}$ pricing strategy pays a total cost of $\Theta(M)$.

Examples of such distributions include point-mass distributions at $p_j > 0$ for each category j (e.g., a collective contract that sets wages in each market segment) or truncated distributions for which there is no worker density below a price floor $p > 0$. Such markets can occur naturally when the workers do not have uncertainty over their costs.

Sublinear wage suppression regime. Empirical research shows that markets in the gig economy are not often coordinated. Without the positive left gap property, we transition to a regime where the total cost will achieve a sublinear bound:

Lemma 3.5 (Sublinear total cost). *If for each $i \in [M]$,*

$$\inf\{x \in [0, 1] : D_i(x) > 0\} = 0, \quad (9)$$

the $\mathbf{p}^{\text{SWS}}$ pricing strategy pays a total cost of $o(M)$.

A large array of distributions fall under this regime. For example, CDFs with polynomially vanishing density at 0 such as a power left tail: $\exists a > 1, K > 0$ s.t. $D(x) = Kx^a(1 + o(1))$ as $x \rightarrow 0$. A specific example is the Beta($a > 1, 1$) distribution, noting that varying a achieves a sublinear bound $\Theta(M^{1-1/a})$ with any exponent (see Observation A.2 in the Appendix for computation and further examples). As such, a positive density at 0 implies a phase transition in the total wages to become sublinear.

Logarithmic wage suppression regime. A further transition occurs when the left tail grows linearly, as a special case of the aforementioned polynomial; in this regime, the stochastic wage suppression strategy achieves a total cost bound of $O(\log(M))$. As described in Section 2, this regime can be exemplified as *high dispersion markets* where workers have high uncertainty over their costs. We state a distribution-agnostic sufficient regularity condition that allows a strategic principal to achieve a cost of $O(\log M)$ while preserving the linear wait time guarantee. Informally, this condition implies a “linear left-tail growth,” i.e., $D_i(x) \gtrsim x$ near 0.

Lemma 3.6 (Logarithmic total cost). *We denote by left-tail regularity the following condition: there exist constants $c > 0$ and $x_0 \in (0, 1]$ such that for all $i \in [M]$ and all $x \in [0, x_0]$,*

$$D_i(x) \geq cx. \quad (10)$$

Equivalently, for all $u \in [0, cx_0]$, $Q_i(u) \leq u/c$, or $\mathbb{P}(v_i < 1/t) = \Omega(1/t)$, $\forall t \in [M]$. For any distributions $\{D_i\}_{i \in [M]}$ with property 10, the $\mathbf{p}^{\text{SWS}}$ pricing strategy pays a total cost of $O(\log(M))$.

Examples of distributions that fall under property 10 are the uniform distribution, the exponential distribution, and any left-skewed power law with a heavy tail, such as Beta(1, b) for $b < 1$ (see Observation A.3 in the Appendix for exact computations).

Constant wage suppression regime. A further phase transition occurs where there is a significant mass concentrated near 0 (so, most workers will have valuations arbitrarily close to 0): in this regime, the principal's pricing strategy suppresses wages down to

a total cost of $O(1)$ with respect to M . This regime is emblematic of distributions exhibiting a *right-skew* described in Section 2, which characterizes *vulnerable markets*. We express a formal criterion below, with a generalization in the Appendix (Proposition A.4).

Lemma 3.7 (Constant total cost). *If distributions D_i have a heavier than linear left tail, such as a polynomial, $\exists x_0 \in (0, 1]$, $b \in (0, 1)$, $c > 0$ s.t. $\forall i \in [M]$, $x \in [0, x_0]$*

$$D_i(x) \geq cx^b. \quad (11)$$

Equivalently, for all $u \in [0, cx_0]$, $Q_i(u) \leq (u/c)^{1/b}$, or $\mathbb{P}(v_i < 1/t) = \Omega(t^{-b})$, $\forall t \in [M]$. For any distributions $\{D_i\}_{i \in [M]}$ with property 11, the $\mathbf{p}^{\text{SWS}}$ pricing strategy pays a total cost of $O(1)$ with respect to M .

In addition, any distribution with an atom at 0 will also satisfy the same properties from Lemma 3.7.

We remark on the phase transition that the principal's strategy achieves: without collective action, pricing at the quantiles obtains sublinear total cost for the principal; depending on the exact distribution family, the total cost can vary from any sublinear bound, down to a logarithmic, and even a constant wage suppression regime as a function of M . These regimes mirror reported empirical properties of workers' valuations distributions in the gig economy. We provide sufficient conditions to characterize the principal's ability to suppress wages in different regimes. Figure 4 presents the total cost that $\mathbf{p}^{\text{SWS}}$ achieves for different regimes (top row, in blue for $\alpha = 0$, equivalent to no collective action). Different wage suppression regimes are exemplified through different distributions: $O(1)$ in panel (a), $O(\log(M))$ in panels (b)-(c) and $O(M^{1/2})$ in panel (d) for the total cost.

Remark 3.8 (A note on cost optimality). *The principal's strategy for quantile pricing is optimal when the valuations distributions are the same across categories. When these distributions come from a heterogeneous family of distributions, the principal may further optimize his cost by exploiting the heterogeneity (e.g., via contextual bandits [35]). Methods for optimizing the absolute cost of the principal are left for future work.*

4 Random collectives fail to prevent wage suppression

In this section, we explore **untargeted** collective action strategies: in these strategies, a random fraction of the workers coordinate on a price floor (similar to the process described in Hardt et al. [25]), essentially only accepting tasks that are offered by the principal above a certain price. If all workers fully coordinate on all categories, that imposes a minimum price floor and thus prevents wage suppression as the total cost is $\Theta(M)$. Here, we are interested in the power of the collective as a function of its size.

Horizontal α -fraction coordination. Let $\alpha \in [0, 1)$ denote the participation rate in the collective. A collective sets a category-specific minimum price $\bar{p}_j, \forall j \in [M]$. Since the workers arrive independently and sequentially, we model their valuations as drawn from the mixture distribution:

$$v_j \sim (1 - \alpha) \cdot D_j + \alpha \cdot \delta_{\bar{p}_j}, \quad (12)$$

where $\delta_{\bar{p}_j}$ is a point mass at the collective-set price $\bar{p}_j \in [0, 1]$. Then, the CDF of this mixture distribution is

$$D_j^{\text{mix}}(x) = (1 - \alpha) \cdot D_j(x) + \alpha \cdot \mathbf{1}\{x \geq \bar{p}_j\}, \quad x \in [0, 1]. \quad (13)$$

We show that under this data-generating distribution, the stochastic wage suppression pricing strategy achieves a wait time for the principal of $O(M)$ with high probability, no matter the size of the collective (the value of α). Essentially, the random collectives *thin* the arrival rate of successful workers, but leave a positive mass of workers who take up a task at low prices which can continue to be exploited by a principal with only a constant factor in the wait time.

Lemma 4.1. *Let $\mathbf{p}_\alpha^{\text{SWS}}$ denote the SWS pricing strategy applied to the mixture distributions $\{D_j^{\text{mix}}\}_{j \in [M]}$. Then*

$$\mathbb{E}[\text{WAIT}(\mathbf{p}_\alpha^{\text{SWS}})] \leq \frac{M}{q_*(\alpha)} = O(M), \quad (14)$$

$$\text{WAIT}(\mathbf{p}_\alpha^{\text{SWS}}) = O\left(M + \sqrt{M \log(1/\delta)} + \log(1/\delta)\right)$$

with probability at least $1 - \delta$, where $q_(\alpha) := 1 - \exp(-(1 - \alpha)\theta)$ and with constants depending only on θ and α .*

PROOF SKETCH. We note that the success probability in each iteration will still be lower bounded by a constant, which now depends on α . This gives the same computation for the expected wait time, scaled by an α -dependent constant, and the concentration inequality follows exactly like in Theorem 3.3. $\square$

Furthermore, the effect of the random collective only affects the principal's strategy by a constant in the total cost for any $\alpha < 1$:

Lemma 4.2. *Fix $\alpha \in [0, 1)$ and $\theta > 0$, and let $\bar{p}^- := \min_{i \in [M]} \bar{p}_i$ denote the minimum collective price floor (assumed positive). Then, across all market regimes analyzed in Section 3,*

$$\text{COST}(\mathbf{p}_\alpha^{\text{SWS}}) = \Theta(\text{COST}(\mathbf{p}^{\text{SWS}})) \quad (15)$$

as a function of the number of categories M .

PROOF SKETCH. The mixture distribution forces the stochastic wage suppression strategy to increase prices, as the quantile values shift: the new quantile values are either the original quantile values of the original distributions D_i shifted by a factor of $1/(1 - \alpha)$ (when the worker is not part of the collective), or equal to the minimal price floor $\bar{p}^-$. In total, this shifts the total cost by a factor of $1/(1 - \alpha)$ that follows from quantile reweighting under the mixture distribution and a constant addition for the collective workers. For the regimes described in Section 3, this translates to $\text{COST}(\mathbf{p}_\alpha^{\text{SWS}}) = \frac{1}{1 - \alpha} \text{COST}(\mathbf{p}^{\text{SWS}}) + O(1)$ (full proof in the Appendix). Thus, the total cost will undergo the same phase transitions as described in the case without any collective action. $\square$

Figure 4 showcases the total cost (top row) and the total wait time (bottom row) for various values of α for the $\mathbf{p}_\alpha^{\text{SWS}}$ pricing strategy. The collective size does not affect the asymptotic growth rate of cost or wait time, but it does change constants (computed in closed form in the Appendix): larger collectives increase both cost and wait time by constant factors.

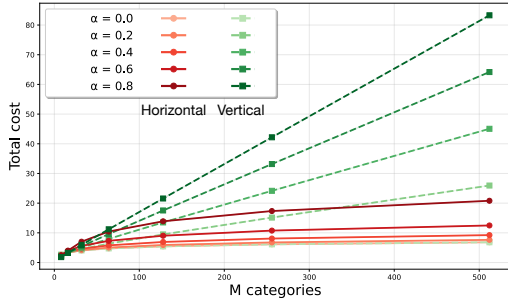


Figure 3: A comparison of the total cost achieved by a horizontal (untargeted) collective vs a vertical (targeted) collective with a budget of $\epsilon = 0.01$ under the p^{SWS} pricing strategy and uniform valuations distributions.

Remark 4.3. We note that this untargeted collective action model is inspired by decentralized organizing mechanisms [25, 54]. Our results show that such uncoordinated efforts cannot overcome structural wage suppression: as long as there remains positive mass in the lower tail of valuations, a principal can continue to exploit low-cost workers. Thus, recruiting anything less than all low-valuation workers preserves our bounds qualitatively, since untargeted coordination primarily thins the arrival rate of low-cost acceptances. For this reason, while information campaigns have been deemed helpful for worker organizing [28, 60], their efficacy may be limited if they fail to reach essentially all lowest-paid workers. In our model, information about a category’s market price may reduce variance yet still leave tail properties that sustain sublinear or constant wage suppression regimes.

5 Targeted collective action as (re)distribution

In contrast to untargeted (random) collectives, **targeted** collectives strategically allocate limited organizing resources (e.g., pointed information campaigns or price-floor agreements) to the lowest quantiles, where wage suppression originates. In our model, this means focusing on α -fraction segments of the market and concentrating a small “budget” ϵ of organizing effort in each by targeting the lowest-cost workers.

We refer to such targeted strategies as *vertical coordination*, where “vertical” means organizing within entire market segments (here, segments correspond to task categories). We draw on classical organizing strategies in collective action: vertical collectives can be more effective in dispersed markets [43], can address fairness concerns in economic models [33], and have been applied as industry-region organizing in countries like Germany [30]. Both recent [16] and longitudinal [38, 40] studies show that the shift from trade-based organizational unionizing to more heterogeneous movements across sectors can reduce bargaining power.

Clearly, vertical coordination places workers in the *coordinated markets* regime: for an α -fraction of categories, the intervention instills a price floor, and thus the principal must pay a total cost at least $\Theta(\alpha M)$ (as property 8 is satisfied for an α -fraction of categories). We consider the fraction α a constant and show that even for small α , Lemma 3.4 applies and we recover the tight linear cost regime. We next quantify the minimum budget required to deploy

such targeted interventions in terms of total variation distance on valuation distributions.

Theorem 5.1. Define the critical budget for targeted collective action as

$$\epsilon^* := \inf\{\epsilon > 0 : \inf_i Q_i(\epsilon) > 0\}. \quad (16)$$

For any $\epsilon > \epsilon^*$, we can construct a new set of valuations distributions $\{D_i^*\}_{i \in [M]}$ such that $\text{TV}(D_i^*, D_i) \leq \epsilon$ for all $i \in [M]$, and under which the stochastic wage suppression pricing strategy must incur total cost

$$\text{COST}(p^{\text{SWS}}) = \Theta(M).$$

PROOF SKETCH. For any $\epsilon \geq \epsilon^*$, define the valuation distributions after targeted collective action for category i by removing ϵ mass from the lowest-cost region and placing it at 1:

$$D_i^*(x) := \max\{D_i(x) - \epsilon, 0\} \text{ for } x < 1, \quad D_i^*(1) = 1. \quad (17)$$

This transformation moves at most ϵ probability mass, hence $\text{TV}(D_i^*, D_i) \leq \epsilon$. We then claim that for every $u \in (0, 1 - \epsilon)$, the quantiles of the adjusted distribution satisfy $Q_i^*(u) \geq Q_i(u + \epsilon)$ (proved in Proposition A.7 in the Appendix). Next, we note that the p^{SWS} strategy satisfies a constant probability of success of at least $1 - e^{-\theta}$ for some $\theta > 0$ in each iteration t , since it is defined as

$$p_t \geq \max_{i \in A_t} Q_i^*\left(\frac{\theta}{M - t + 1}\right) \geq \max_{i \in A_t} Q_i\left(\epsilon + \frac{\theta}{M - t + 1}\right) \geq \max_{i \in A_t} Q_i(\epsilon). \quad (18)$$

Hence the total payment obeys

$$\begin{aligned} \text{COST}(p^{\text{SWS}}) &= \sum_{t=1}^M p_t \geq \sum_{t=1}^M \max_{i \in A_t} Q_i(\epsilon) \geq M \cdot \underline{q}_\epsilon, \\ \text{where } \underline{q}_\epsilon &:= \min_{i \in [M]} Q_i(\epsilon). \end{aligned} \quad (19)$$

In particular, as long as $\underline{q}_\epsilon > 0$ (a constant independent of M), the principal’s cost is $\Theta(M)$, with a wait time of $O(M)$ with constants depending on ϵ . Without this condition, the principal can still exploit tail behavior and achieve sublinear total cost: we need that the lowest ϵ -mass of the distribution is not “free”, so the workers in this region do not take arbitrarily low payments. In that case, we need a critical budget high enough to cover the entire mass. $\square$

Remark 5.2. For clarity, in Theorem 5.1 the intervention is applied to all categories; applying it to an α -fraction forces cost $\Theta(\alpha M)$. We connect Theorem 5.1 with different market regimes in terms of the critical budget necessary to prevent wage suppression:

- **No campaign is needed in coordinated markets:** in markets where there is already a positive price floor (property (8)), collective action is not necessary, since the total cost is $\Theta(M)$.
- **When any targeted campaign prevents wage suppression:** if targeted categories have no atom at 0 (and support starts above 0 after any ϵ shift), then $\epsilon^* = 0$ for those categories. This can restore any sublinear wage suppression regime to a linear one, including logarithmic and constant. This is significant as it includes a wide range of distributions, as described in Section 3, and hence various market conditions.

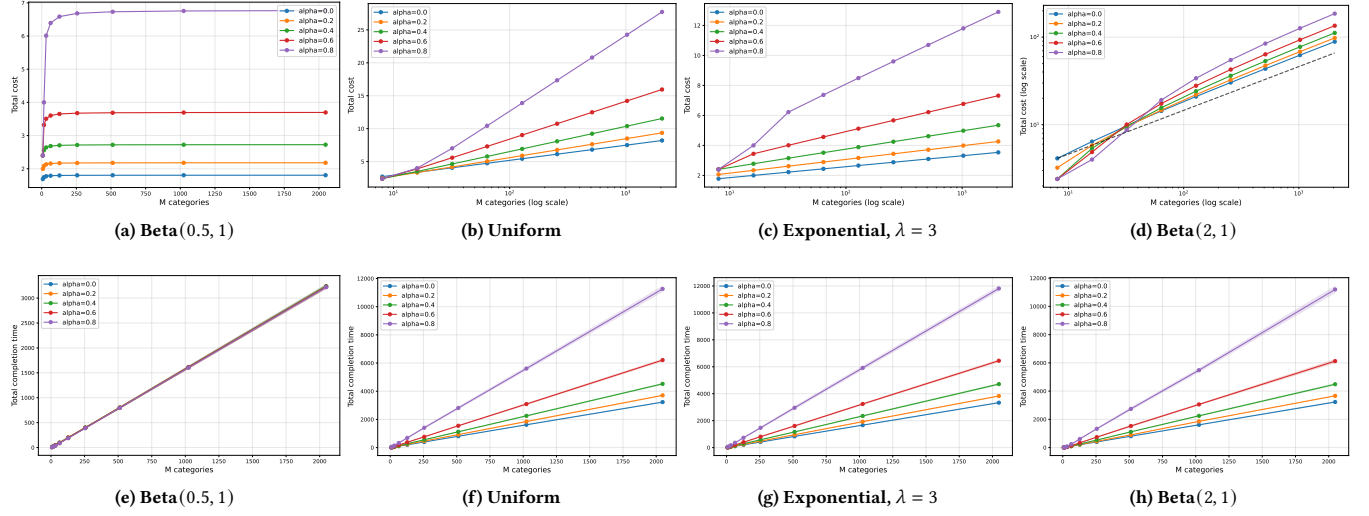


Figure 4: Total cost (top row) and total wait time (bottom row) under p^{SWS} for various values of α and M , for a collective price floor value of 0.2 and different workers' valuations distributions. Panel (a) highlights the $O(1)$ total cost regime, panels (b)–(c) highlight the $O(\log(M))$ total cost regime in log scale, and panel (d) highlights a $O(M^{1/2})$ regime of total cost in log-log scale (dashed line represents the 0.5 slope line). The total wait time is $O(M)$ with constants depending on α and distribution families.

- **When a targeted campaign needs a minimum budget:** if more than $(1 - \alpha)$ -fraction of the valuations distributions have an atom at 0, i.e. $\epsilon^* = \rho$ for some $\rho > 0$, that means that a targeted collective must ‘move’ the entire atom in order to shift the total price from $O(1)$ to $O(M)$. This implies a choice on the side of the decision-maker: choose a target fraction α that allows targeting distributions without this property, or choose a high enough budget ϵ to cover the mass at 0. Our model allows a decision-maker to make this choice by providing a guarantee of when targeting works and what budget it requires.³

This shows the **power of targeted recruiting** in collective action: horizontal collectives preserve wage suppression, whereas targeted collectives restore the cost bound to $\Theta(M)$ for any small budget as long as an α -fraction of categories do not have an atom at 0. Instead of thinning arrivals, they lift the free-labor region. When such regions have significant mass, collectives are still effective as long as the targeting budget exceeds the atom size. This difference captures the classic tension between breadth and focus in collective action: random collectives diffuse effort across many categories, while targeted collectives spend their limited ϵ budget where it matters most—on the lowest quantiles that cause wage suppression.

Figure 3 illustrates the asymptotic difference between the power of a collective in horizontal versus vertical coordination: for any fixed $\alpha < 1$, a principal can obtain the $O(M)$ wait time guarantee with the stochastic wage suppression strategy in horizontal coordination (in red), essentially keeping the total cost of order $O(\log(M))$. However, vertical coordination (in green) quickly forces a wage increase as M increases, even for small collective sizes and a small budget ϵ for targeting, restoring a total cost of $\Theta(M)$.

³Here, we consider α as an *a priori* chosen collective fraction. The problem of dynamically adjusting α with M is left for future work.

6 Discussion

Our work has implications for both platform designers and worker advocates. On the one hand, it provides a versatile theoretical model for diagnosing when algorithmic pricing can lead to wage suppression, based on the structure of workers' perceived cost distributions. On the other hand, it offers concrete guidance for designing collective interventions that counteract procurement power, highlighting when and why targeted organizing can be effective.

Several open directions emerge. First, our pricing strategy could be extended to dynamic pricing models that learn workers' valuation distributions online from task uptake. While the stochastic wage suppression strategy we analyze is not intended to be cost-optimal, it provides a natural benchmark under partial distributional knowledge and reveals sharp phase transitions in workers' wages. Second, future work could evaluate platform-side interventions that reduce workers' uncertainty about their costs and study how these interact with different models of collective action, including alternative targeting and redistribution mechanisms. Third, an important empirical direction is to characterize valuation distributions in real-world platforms and to examine how they evolve under sustained pricing strategies and organizing efforts.

Overall, our results underscore a structural asymmetry in digital labor markets: while random, untargeted coordination tends to thin participation without altering wage suppression regimes, even modestly targeted collective action can fundamentally change market outcomes.

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A Detailed proofs

We detail proofs of the main results presented in the main text.

A.1 Wait time of SWS

We start with basic properties of the wait time and success probability at iteration t : since workers are drawn sequentially until the first success occurs, $N_t \sim \text{Geom}(q_t)$, with mean and variance

$$\mathbb{E}[N_t] = \frac{1}{q_t}, \quad \text{Var}(N_t) = \frac{1 - q_t}{q_t^2}. \quad (20)$$

As defined before, the total number of workers needed across all iterations is equal to the total wait time $\text{WAIT}(\mathbf{p}^{\text{SWS}}) := \sum_{t=1}^M N_t$. Clearly, $N_t \perp N_{t+1}$, so

$$\mathbb{E}[\text{WAIT}(\mathbf{p}^{\text{SWS}})] = \sum_{t=1}^M \frac{1}{q_t}, \quad \text{Var}(\text{WAIT}(\mathbf{p}^{\text{SWS}})) = \sum_{t=1}^M \frac{1 - q_t}{q_t^2}. \quad (21)$$

As part of our proofs, we also need a standard result on sub-exponential behavior of Geometric variables:

Lemma A.1 (Sub-exponential property of Geometric random variables). *If $\Delta \sim \text{Geom}(q)$ on $\{1, 2, \dots\}$, then $\mathbb{E}\Delta = 1/q$, $\text{Var}(\Delta) = (1 - q)/q^2$. Moreover, there exists a universal $C > 0$ such that*

$$\|\Delta - \mathbb{E}\Delta\|_{\psi_1} \leq \frac{C}{q}, \quad (22)$$

i.e., $\Delta - \mathbb{E}\Delta$ is sub-exponential with scale $O(1/q)$.

PROOF. The mgf of Δ is $M_\Delta(\lambda) = \frac{qe^\lambda}{1 - (1-q)e^\lambda}$ for $\lambda < -\ln(1 - q)$. For $|\lambda| \leq c/q$ with a small enough absolute $c > 0$, a Taylor expansion of $\log M_\Delta(\lambda) - \lambda \mathbb{E}\Delta$ yields $\log \mathbb{E}e^{\lambda(\Delta - \mathbb{E}\Delta)} \leq C_1 \lambda^2 / q^2$, which is the standard Orlicz ψ_1 criterion with scale C/q . $\square$

PROOF OF THEOREM 3.3. By Lemma 3.2, we have $q_t \geq q_* := 1 - e^{-\theta}$ for all $t \in [M]$. Since $N_t \sim \text{Geom}(q_t)$ and the N_t are independent across t ,

$$\mathbb{E}[\text{WAIT}(\mathbf{p}^{\text{SWS}})] = \sum_{t=1}^M \mathbb{E}[N_t] = \sum_{t=1}^M \frac{1}{q_t} \leq \frac{M}{q_*}.$$

We also note that there exists a constant $0 < q_- < 1$ (depending only on θ) such that

$$q_- \leq q_t \quad \text{for all } t \in \{1, \dots, M\}. \quad (23)$$

as per Lemma 3.2 as q^* . As such, we can also compute the variance of the wait time under the pricing strategy $\mathbf{p}^{\text{SWS}}$ as

$$\text{Var}(\text{WAIT}(\mathbf{p}^{\text{SWS}})) \leq \sum_{t=1}^M \frac{1 - q_-}{q_-^2} = O(M). \quad (24)$$

For the concentration bound, apply Lemma A.1 with $q = q_t \geq q_*$ to find ψ_1 control $\|N_t - \mathbb{E}N_t\|_{\psi_1} \leq C/q_*$ for some constant C . Since all N_t are independent, we can apply Bernstein's inequality for the sum of independent sub-exponential variables $X_t = N_t - \mathbb{E}[N_t]$, which sum up to $\text{WAIT}(\mathbf{p}^{\text{SWS}})$.

We thus find that exists a constant $c = c(\theta) > 0$ s.t. $\forall x > 0$,

$$\mathbb{P}(\text{WAIT}(\mathbf{p}^{\text{SWS}}) - \mathbb{E}[\text{WAIT}(\mathbf{p}^{\text{SWS}})] \geq x) \leq \exp\left(-c \min\{x^2/M, x\}\right), \quad (25)$$

or, equivalently, $\exists c^* > 0$ that depends only on c s.t. $\forall \epsilon \in (0, 1)$,

$$\Pr\left(\left|\frac{\text{WAIT}(\mathbf{p}^{\text{SWS}})}{\mathbb{E}[\text{WAIT}(\mathbf{p}^{\text{SWS}})]} - 1\right| \geq \epsilon\right) \leq 2 \exp(-c^* \epsilon^2 M). \quad (26)$$

$\square$

We note that the wait time concentration bound is distribution-agnostic and holds for any D_i under the independence assumptions of the model.

Table 1: Left-tail properties of D and implications for the total cost achieved by $\mathbf{p}^{\text{SWS}}$.

Left-tail class	Total cost under $\mathbf{p}^{\text{SWS}}$
Gap ($g > 0$)	$\Theta(M)$
$D(x) \asymp x$	$\Theta(\log M)$
$D(x) \asymp x^\alpha$, $\alpha > 1$	$\Theta(M^{1-1/\alpha})$
$D(x) \gtrsim x^\beta$, $\beta < 1$	$O(1)$
Atom at 0	$O(1)$

A.2 Total cost of SWS

See Table 1 for different left-tail properties of D .⁴

PROOF OF LEMMA 3.4. We show that the stochastic wage suppression strategy satisfies

$$M g_* \leq \text{COST}(\mathbf{p}^{\text{SWS}}) \leq M \cdot \max_{i \in [M]} Q_i(\theta) \leq M, \quad (27)$$

and thus $\text{COST}(p) = \Theta(M)$ (with constants depending only on g_* and θ). If $D_i(x) = 0$ for all $x < g_i$, then $Q_i(u) \geq g_i$ for every $u \in (0, 1]$. Thus for every iteration t , $p_t = \max_{i \in A_t} Q_i(\theta/(M - t + 1)) \geq \max_{i \in A_t} g_i \geq g_*$, so $\sum_t p_t \geq M g_*$. For the upper bound, use the monotonicity of Q_i in its argument: $Q_i(\theta/(M - t + 1)) \leq Q_i(\theta) \leq 1$, hence $p_t \leq \max_i Q_i(\theta) \leq 1$ and the sum is at most M . $\square$

We note that to achieve $\Theta(M)$, only a fraction of categories need a gap: if αM categories (for some constant $\alpha \in (0, 1]$) satisfy $g_i \geq g_0 > 0$, then $\text{COST}(\mathbf{p}^{\text{SWS}}) \geq \alpha g_0 M$ and still $\text{COST}(\mathbf{p}^{\text{SWS}}) \leq M$.

⁴In Table 1, the $\asymp$ notation means that if $f(x) \asymp g(x)$, then $\exists c_1, c_2 > 0$ s.t. $c_1 g(x) \leq f(x) \leq c_2 g(x)$ for small x ; the $\gtrsim$ notation means that if $f(x) \gtrsim g(x)$, then $\exists c > 0$ s.t. $f(x) \geq c g(x)$ for small x .

PROOF OF LEMMA 3.5. Fix $\varepsilon > 0$. Since $\inf\{x : D_i(x) > 0\} = 0$, we have $Q_i(u) \rightarrow 0$ as $u \downarrow 0$ for every i . Hence, for each i there exists $\delta_i > 0$ such that $Q_i(u) \leq \varepsilon$ for all $u \in (0, \delta_i]$. Let $\delta := \min_{i \in [M]} \delta_i > 0$. Let $J := \lceil \theta/\delta \rceil$. For all $j \geq J$, we have $\theta/j \leq \delta$, and therefore

$$Q_{\max}(\theta/j) = \max_i Q_i(\theta/j) \leq \varepsilon.$$

Writing the total cost as $\sum_{j=1}^M Q_{\max}(\theta/j)$ (with $j = M - t + 1$), we obtain

$$\text{COST}(\mathbf{p}^{\text{SWS}}) \leq \sum_{j=1}^{J-1} Q_{\max}(\theta/j) + \sum_{j=J}^M \varepsilon \leq C(\varepsilon) + \varepsilon M,$$

where $C(\varepsilon) := \sum_{j=1}^{J-1} Q_{\max}(\theta/j)$ is finite and independent of M . Dividing by M and taking $M \rightarrow \infty$ yields $\limsup_{M \rightarrow \infty} \text{COST}(\mathbf{p}^{\text{SWS}})/M \leq \varepsilon$. Since ε was arbitrary, $\text{COST}(\mathbf{p}^{\text{SWS}}) = o(M)$. $\square$

Observation A.2. Canonical examples of distributions with property 9:

- **Beta**($a, 1$) on $[0, 1]$ with $a > 1$ (exactly $D(x) = x^a$) gives a total cost of $\text{COST}(\mathbf{p}^{\text{SWS}}) = \Theta(M^{1-1/a})$. Similarly, the **Gamma**($k > 1, \lambda$) on $[0, \infty)$ with $k > 1$ and the **Weibull**($k > 1, \lambda$) distributions reduce to the same computations.
- **Weibull-at-zero type** distribution (stretched-exponential left tails). Take a CDF of the form $D(x) = \exp(-\lambda x^{-\beta} (1+o(1)))$ as $x \rightarrow 0$, for $\beta, \lambda > 0$. Then

$$Q(u) = \left(\frac{\lambda}{\ln(1/u)} \right)^{1/\beta} (1+o(1)), \quad (28)$$

giving $\text{COST}(\mathbf{p}^{\text{SWS}}) = \Theta\left(\frac{M}{(\ln M)^{1/\beta}}\right)$. This is sublinear but can get arbitrarily close to linear: for any fixed $\varepsilon > 0$ and large M , $M/(\ln M)^{1/\beta} \gg M^{1-\varepsilon}$.

- **Lognormal left tail.** Take the $\text{LogNormal}(\mu, \sigma^2)$ with a super-light tail. For small u ,

$$Q(u) = \exp\left(\mu + \sigma \Phi^{-1}(u)\right) = \exp\left(\mu - \sigma \sqrt{2 \ln(1/u)} (1+o(1))\right). \quad (29)$$

With $u = \theta/|A_t|$,

$$Q(\theta/|A_t|) = \exp\left(\mu - \sigma \sqrt{2 \ln |A_t|} (1+o(1))\right), \quad (30)$$

Taking the integral gives a total cost bound of

$$\text{COST}(\mathbf{p}^{\text{SWS}}) = \sum_{j=1}^M Q(\theta/j) = \Theta\left(M \exp(-\sigma \sqrt{2 \ln M})\right). \quad (31)$$

PROOF OF LEMMA 3.6. Property 10 implies that there are constants $c, t_0 > 0$ such that for all $t > t_0$ and all category indices i , $\mathbb{P}[v_i \leq 1/t] \geq c/t$. Thus, for $x \leq 1/t_0$, $D_i(x) \geq cx$, or equivalently, $Q_i(u) \leq u/c$. Thus, we can split the total cost in two sets: a set of $j \geq \lceil \theta t_0/c \rceil$ for which $p_t \leq \frac{\theta}{cj}$ and a set $j \leq \lceil \theta t_0/c \rceil$ which is finite and constant (denoted by C) with respect to M . Then, the total cost can be bounded as

$$\text{COST}(\mathbf{p}^{\text{SWS}}) = \sum_{t=1}^M p_t \leq C + \sum_{j=\lceil \theta t_0/c \rceil}^M \frac{\theta}{cj} = O(\log(M)) \quad (32)$$

$\square$

Observation A.3. Canonical examples of distributions with property 10 (note that any log-concave law with pdf $f(0) > 0$ will satisfy this bound):

- **Uniform** $[0, 1]$. $Q(u) = u$, so $p_t = \theta/(M - t + 1)$ and $\text{COST} = \theta \sum_{j=1}^M \frac{1}{j} = \Theta(\log M)$.
- **Exponential** (λ) **truncated to** $[0, 1]$. Near 0, $D(x) \sim \lambda x$, hence property 10 holds with $c = \lambda/2$ for small enough x_0 , so $\text{COST}(\mathbf{p}^{\text{SWS}}) = O(\log M)$. A direct calculation of the quantiles of $D(x) = 1 - e^{-\lambda x}$ gives $Q(1/t) = -1/\lambda \ln(1 - 1/t) = \Theta(1/t)$ through a simple Taylor expansion, thus summing to $\text{COST}(\mathbf{p}^{\text{SWS}}) = \Theta(\log M)$.
- **Beta**($1, b$) with $b < 1$: we can compute the quantiles as $Q(1/t) = 1 - (1 - 1/t)^{1/b} = \Theta(1/t)$ by a similar Taylor expansion, obtaining again $\text{COST}(\mathbf{p}^{\text{SWS}}) = \Theta(\log M)$.

We formulate a general criterion for the total price to stay constant with respect to M :

Proposition A.4. The total cost of the principal is $O(1)$ with respect to M if there exist constants $c_1, c_2 > 0$ such that for all $M \geq 2$,

$$\begin{aligned} c_1 \int_1^M Q_{\max}\left(\frac{\theta}{x}\right) dx &\leq \sum_{j=1}^M Q_{\max}\left(\frac{\theta}{j}\right) \\ &\leq c_2 \left(Q_{\max}(\theta) + \int_1^M Q_{\max}\left(\frac{\theta}{x}\right) dx \right), \end{aligned} \quad (33)$$

for $Q_{\max}(u) = \max_i Q_i(u)$. Equivalently, with the change of variables $u = \theta/x$,

$$\int_1^M Q_{\max}\left(\frac{\theta}{x}\right) dx = \theta \int_{\theta/M}^{\theta} \frac{Q_{\max}(u)}{u^2} du. \quad (34)$$

The total cost is finite if and only if $\int_{\theta/M}^{\theta} \frac{Q_{\max}(u)}{u^2} du < \infty$.

PROOF. The proposition follows from the nondecreasing property of $Q_{\max}(u)$ and applying the integral test for monotone sequences to $Q_{\max}(\theta/x)$ with the mentioned change of variables. $\square$

PROOF OF LEMMA 3.7. If there exist $x_0 \in (0, 1]$, $\beta \in (0, 1)$ and $c > 0$ such that for all i and $x \in [0, x_0]$, $D_i(x) \geq cx^\beta$, we can compute the quantiles for all small u , $Q_i(u) \leq (u/c)^{1/\beta}$. Thus,

$$p_t \leq Q_{\max}(\theta/j) \leq (\theta/c)^{1/\beta} j^{-1/\beta}. \quad (35)$$

Since $1/\beta > 1$, $\sum_{j \geq 1} j^{-1/\beta} < \infty$, $\sum_t p_t = O(1)$ w.r.t. M . Finally, if the distributions D_i have an atom at 0, it means that $D_i(0) \geq \rho_i > 0$, then for $j \geq \theta/\rho_i$, $Q_i(\theta/j) \leq 0$. Given nonnegative prices, only finitely many terms are positive, so $\sum_t p_t = O(1)$ w.r.t. M . $\square$

Observation A.5 (Phase transition behavior for the Beta($a, 1$) distribution family). The Beta distribution can be used to illustrate all regimes of growth for the total price of the principal under the stochastic wage suppression pricing strategy discussed in the main text. If $D(x) = x^a$ (i.e., $X \sim \text{Beta}(a, 1)$), then $Q(u) = u^{1/a}$, so $p_t = (\theta/(M - t + 1))^{1/a}$. Then

$$\text{COST}(\mathbf{p}^{\text{SWS}}) = \sum_{j=1}^M \left(\frac{\theta}{j}\right)^{1/a} = \begin{cases} \Theta(M^{1-1/a}), & a > 1, \\ \Theta(\log M), & a = 1, \\ \Theta(1), & a < 1. \end{cases} \quad (36)$$

PROOF. The quantiles of the Beta distribution can be computed in closed form as $Q(u) = u^{1/\alpha}$. Then, the standard asymptotics of $\sum_{j=1}^M j^{-\alpha}$ apply: it diverges like $M^{1-\alpha}$ for $\alpha \in (0, 1)$, like $\log M$ for $\alpha = 1$, and converges for $\alpha > 1$. $\square$

Observation A.6 (On optimality under distributional knowledge.). *Given full knowledge of $\{D_i\}_{i \in A_t}$, the minimum price at iteration t necessary to satisfy the iteration-level success floor q_* is*

$$p_t^* \in \arg \min_{p \in [0,1]} \left\{ p : 1 - \prod_{i \in A_t} (1 - D_i(p)) \geq q_* \right\}. \quad (37)$$

The pricing strategy based on the $1/|A_t|$ quantiles essentially assumes that the principal has partial knowledge, through an oracle access to the $\{Q_i(u) : i \in A_t\}$ at $u = 1/|A_t|$.

A.3 Random collective action

PROOF OF LEMMA 4.1. For clarity, we present the calculation for the uniform family; the same argument extends to distributions satisfying left-tail regularity $D_i(x) \geq cx$. We compute at iteration t and for an active category i ,

$$D_i^{\text{mix}}(p_t) \geq (1 - \alpha) p_t, \quad (38)$$

hence the probability of success in iteration t can be written as

$$\begin{aligned} q_t &= 1 - \prod_{i \in A_t} (1 - D_i^{\text{mix}}(p_t)) \geq 1 - \left(1 - (1 - \alpha) p_t\right)^{|A_t|} \\ &\geq 1 - \exp(-(1 - \alpha) \theta) =: q_*(\alpha). \end{aligned} \quad (39)$$

Thus, we obtain again that $\Delta_t \sim \text{Geom}(q_t)$ with $q_t \geq q_*(\alpha)$, and thus

$$\mathbb{E}[\text{WAIT}(\mathbf{p}_\alpha^{\text{SWS}})] \leq \frac{M}{q_*(\alpha)} = O(M). \quad (40)$$

The concentration proof around $\mathbb{E}[\text{WAIT}(\mathbf{p}_\alpha^{\text{SWS}})]$ follows exactly like in the proof of Theorem 3.3: for any x , $D_i(x) \geq (1 - \alpha)x$ because the uniform component contributes $(1 - \alpha)x$ and the point mass only increases the CDF. We plug $x = p_t$ and note that the success probabilities are still geometric due to the independence of the N_t random variables. The rest of the proof is identical to Theorem 3.3, as the tail bound follows from applying Bernstein's inequality for the sum of independent sub-exponential variables, with q_* replaced by $q_*(\alpha)$. $\square$

PROOF OF LEMMA 4.2. Let $p^- := \bar{p}_t$ and let $D_i(p^-)$ denote the left limit at p^- . For the mixture CDF $D_i^{\text{mix}}(x) = (1 - \alpha)D_i(x) + \alpha 1\{x \geq p^-\}$, the left-quantile function Q_i^α satisfies

$$Q_i^\alpha(u) = \begin{cases} Q_i\left(\frac{u}{1-\alpha}\right), & 0 \leq u < (1 - \alpha)D_i(p^-), \\ p^-, & (1 - \alpha)D_i(p^-) \leq u \leq \alpha + (1 - \alpha)D_i(p^-), \\ Q_i\left(\frac{u - \alpha}{1 - \alpha}\right), & \alpha + (1 - \alpha)D_i(p^-) < u \leq 1. \end{cases}$$

We partition the sum of the quantiles in three parts over the quantiles $\theta/|A_t|$, a first threshold $u_1 = \lceil \theta/(1 - \alpha)D_i(p^-) \rceil$ and a second threshold $u_2 = \lfloor \theta/(\alpha + (1 - \alpha)D_i(p^-)) \rfloor$ for the index set $1, 2, \dots, M$. These thresholds are independent of M , and thus we have a constant number of them in the price floor regime p^- and in the upper regime, whereas the lower quantile regime is just shifted by a factor of $1/(1 - \alpha)$ from the sum in the original distribution

D_i . We apply the same for taking the maximum over the quantiles in the computation of $\mathbf{p}_\alpha^{\text{SWS}}$ evaluated at each quantile. As such, for the distribution regimes we have analyzed in closed-form as a function M , there exists constants $C_1, C_2 > 0$ s.t.

$$\frac{1}{1 - \alpha} \text{COST}(\mathbf{p}^{\text{SWS}}) - C_1 \leq \text{COST}(\mathbf{p}_\alpha^{\text{SWS}}) \leq \frac{1}{1 - \alpha} \text{COST}(\mathbf{p}^{\text{SWS}}) + C_2 \quad (41)$$

This is because for any distributions with properties 9, 10, or 11, we can use a change of variable $u \rightarrow u/(1 - \alpha)$. In the case where all distributions D_i are the same with quantiles $Q_i := Q$, it is trivial to notice that $\text{COST}(\mathbf{p}_\alpha^{\text{SWS}}) = \sum_{j=1}^M Q^\alpha\left(\frac{\theta(1-\alpha)}{|A_t|}\right)$ is approximately $\frac{1}{1-\alpha} \int_{1-\alpha}^{M(1-\alpha)} Q(\theta/x) dx$. The same argument follows for different distributions D_i that come from the same category (or families with the same tail properties). For distributions that have a positive gap at 0, collective action is not actually needed, since the total cost paid by the principal is $\Theta(M)$ in the absence of any collective. $\square$

A.4 Targeted collective action

Proposition A.7. *For every $u \in (0, 1 - \epsilon)$, the quantiles of the adjusted distribution*

$$D_i^*(x) := \max\{D_i(x) - \epsilon, 0\} \text{ for } x < 1, \quad D_i^*(1) = 1. \quad (42)$$

satisfy

$$Q_i^*(u) \geq Q_i(u + \epsilon) \quad (43)$$

for any $i \in [M]$.

PROOF. For an x that satisfies $D_i^*(x) \geq u > 0$, it means that $\max(D_i(x) - \epsilon, 0) = D_i(x) - \epsilon$, so $D_i^*(x) = D_i(x) - \epsilon \geq u$, so $D_i(x) \geq \epsilon + u$. Coming back to the definition of the quantiles and taking infimums over x , we get equation 43. The inequality does not hold for $u = 0$, but that case is not needed in our set-up since our quantile prices are set for non-negative quantiles. Note that for $u \geq 1 - \epsilon$, both quantiles are 1: $Q_i^*(u) = Q_i(u + \epsilon) = Q_i(1)$, since the distributions have support on $[0, 1]$. This holds for any $i \in [M]$.

Then, we can show that $\text{TV}(D_i^*, D_i) \leq \epsilon$ for all i . To see this, take an arbitrary category $i \in [M]$ and denote by μ_i the measure on $[0, 1]$ associated to the CDF D_i . We assume that $\epsilon < \mu([0, 1])$ (so we shave off less than the available total mass). By construction, we had removed the lowest ϵ mass, but this holds for any measure with mass ϵ , so let's denote by l a measure on $[0, 1]$ with $l([0, 1]) = \epsilon$. Then, we can define another measure ν as

$$\nu = \mu - l + \epsilon \delta_1, \quad (44)$$

where δ_1 is just the Dirac function at 1 (where we placed the ϵ mass). Then, we associated D_i^* with ν and see that $\text{TV}(D_i^*, D_i) = \epsilon$. By the total variation distance,

$$\text{TV}(D_i^*, D_i) = \sup_{A \subseteq [0,1]} |\mu(A) - \nu(A)| \quad (45)$$

For any Borel set A , $\mu(A) - \nu(A) = l(A) - \epsilon \delta_1(A)$. If $1 \notin A$, this is equal to $l(A) \leq l([0, 1]) = \epsilon$ since $A \subseteq [0, 1]$. If $1 \in A$, then its absolute value is less than $\max(l(A) - \epsilon, \epsilon - l(A)) \leq \epsilon$ as above. Finally, take $A = [0, 1]$ to see that $\mu(A) - \nu(A) = l([0, 1]) = \epsilon$, hence $\text{TV}(D_i^*, D_i) = \epsilon$. $\square$